1] Find the sending impedance of the line having negligible losses, where Z_0 is $55\,\Omega$, receiving end impedance $155 + j75\,\Omega$ and line is 1.183 times of wavelength.

$Z_0 = 55\,\Omega$

$Z_R = 155 + j75\,\Omega$

$l = 1.183\lambda$

wkt $\lambda = \frac{2\pi}{\beta}$

$\beta = \frac{2\pi}{\lambda}$

$l = 1.183\lambda$

$\beta l = \frac{2\pi}{\cancel{\lambda}}(1.183)\cancel{\lambda}$

$\beta l = 2\pi(1.183) = (2 \times 3.14 \times 1.183)$

$\beta l = 7.429 \Rightarrow \boxed{\beta l = 7.43}$

$$= 55\left[\frac{(155 + j75) + j(55)\tan(7.43)}{55 + j(155 + j75)\tan(7.43)}\right]$$

$$= 55\left[\frac{(155 + j75) + j(55)(2.29)}{55 + (155j - 75)(2.29)}\right]$$

$$= 55\left[\frac{155 + j75 + 122.045j}{55 + 343.94j - 166.425}\right]$$

$$= 55\left[\frac{155 + 197.045j}{-111.425 + 343.945j}\right]$$

$$= 55 \left[\frac{250 \cdot 700 \angle 51 \cdot 811}{361 \cdot 54 \angle 107 \cdot 950} \right]$$

$$= 55 \cdot \left[0 \cdot 693 \angle -56 \cdot 139 \right]$$

$$\boxed{\begin{aligned} Z_{in} &= 38 \cdot 115 \angle -56 \cdot 139 \ (\Omega) \\ Z_{in} &= 21 \cdot 237 - 31 \cdot 650 j \end{aligned}}$$

2] A transmission lines of characteristics impedance 600 Ω is terminated by a reactance of j150 Ω. Find input impedance of a section 25cm long at a frequency of 300 MHz.

Sol: given, $Z_0 = 600\ \Omega$

$f = 300$ MHz

$l = 25$ cm $\Rightarrow l = 25 \times 10^{-2}$ m

$Z_r = j150$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6}$$

$$= \frac{300 \times \cancel{10^6}}{300 \times \cancel{10^6}}$$

$$\boxed{\lambda = 1}$$

$$\beta = \frac{2\pi}{\lambda} = 6 \cdot 28$$

$$\beta l = 6 \cdot 28 \times 25 \times 10^{-2}$$

$$= 1 \cdot 57 \times 57 \cdot 3$$

$$\beta l = 89 \cdot 961$$

(5)

$$Z_{in} = Z_0 \left[\frac{Z_R + jZ_0 \tan(\beta l)}{Z_0 + jZ_R \tan(\beta l)} \right]$$

$$= 600 \left[\frac{j150 + j600 \times 89.961}{600 + (-150) \times 89.961} \right]$$

$$= 600 \left[\frac{54126.6j}{-40924.15} \right]$$

$$= 600 [-3.4403j]$$

$$\boxed{Z_{in} = 2304.2\ \Omega.}$$

3] A transmission line with a 'Z_0' of 300 Ω is terminated in a purely resistive load. It is found by measurement and the maximum value is 7.5μV. Calculate the value of load resistance.

Given, $Z_0 = 300\ \Omega$

$V_{max} = 7.5\mu V$

$V_{min} = 5\mu V$

Soln: The standing wave ratio is,

$$S = \left| \frac{V_{max}}{V_{min}} \right| = \frac{7.5 \times 10^{-6}}{5 \times 10^{-6}}$$

$$\boxed{S = 1.5}$$

SWR with relation of 'k' is,

$$k = \frac{S-1}{S+1} = \frac{1.5-1}{1.5+1}$$

$$= \frac{0.5}{2.5}$$

$$\boxed{k = 0.33}$$

W.K.T,

$$k = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$0.33 = \frac{Z_R - 300}{Z_R + 300}$$

$$Z_R - 300 = 0.33 Z_R + 99$$

$$Z_R - 0.33 Z_R = 399$$

$$Z_R (1 - 0.33) = 399$$

$$\boxed{Z_R = 595.522\ \Omega.}$$

4] Determine the following is i] standing wave ratio ii] load admittance iii] Impedance of transmission line at the maximum and minimum of the stationary waves along the ~~liki~~ line. The Distance between the load and the first voltage maximum for the transmission line with characteristic impedance of 50 Ω with a receiving end impedance of 100 + 121j Ω the

wavelength of the electrical signal along the line is 2.5m.

Given. $Z_0 = 50\,\Omega$

$Z_R = 100 + 121j$

$\lambda = 2.5m$

Soln: To find Normalized load impedance.

$$Z_n = \frac{Z_R}{Z_0} = \frac{100+121j}{50} = \frac{156.97\angle 50.43}{50}$$

$$= 3.14\angle 50.43$$

$$Z_n = 2.0 + 2.4j$$

i] $S = 5.2$

ii] Load Admittance

$$Y_n = \frac{1}{Z_n} = \frac{1}{2+2.4j} = \frac{1}{3.12\angle 50.19} = 0.32\angle -50.19$$

$$Y_n = 0.2 - 0.25j$$

$$Y_R = \frac{Y_n}{Z_0} = \frac{0.2-0.26j}{50} = \frac{0.33\angle -52.43}{50}$$

$$= 6.6\times 10^{-3}\angle -52.43$$

$$= 4.02\times 10^{-3} - 5.2\times 10^{-3}j$$

$$= 0.004 - j\,0.0052$$

iii] $Z_{max} = SWR = 5.2 \Rightarrow Z_{max} = Z_0 Z_{max}$ (or) $Z_0(S)$

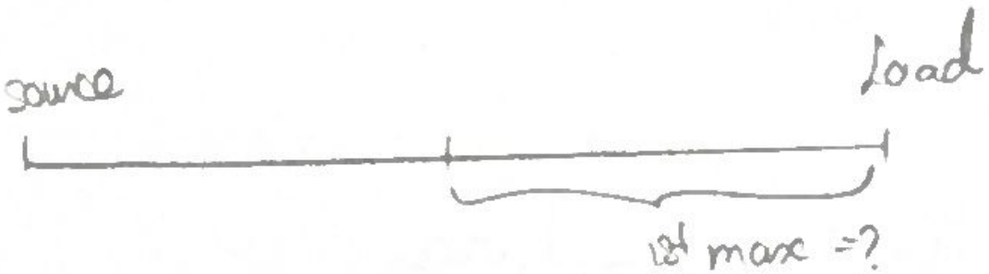
$$= 50(5.2) = 260\,\Omega$$

$$I_{min} = \frac{1}{SWR} = \frac{1}{5.2} = 0.19$$

$$I_{MIN} = I_0 I_{min} \text{ (or) } \frac{I_0}{S}$$

$$I_{MIN} = 50\,(0.19) = 9.5\,\Omega$$

iv] .



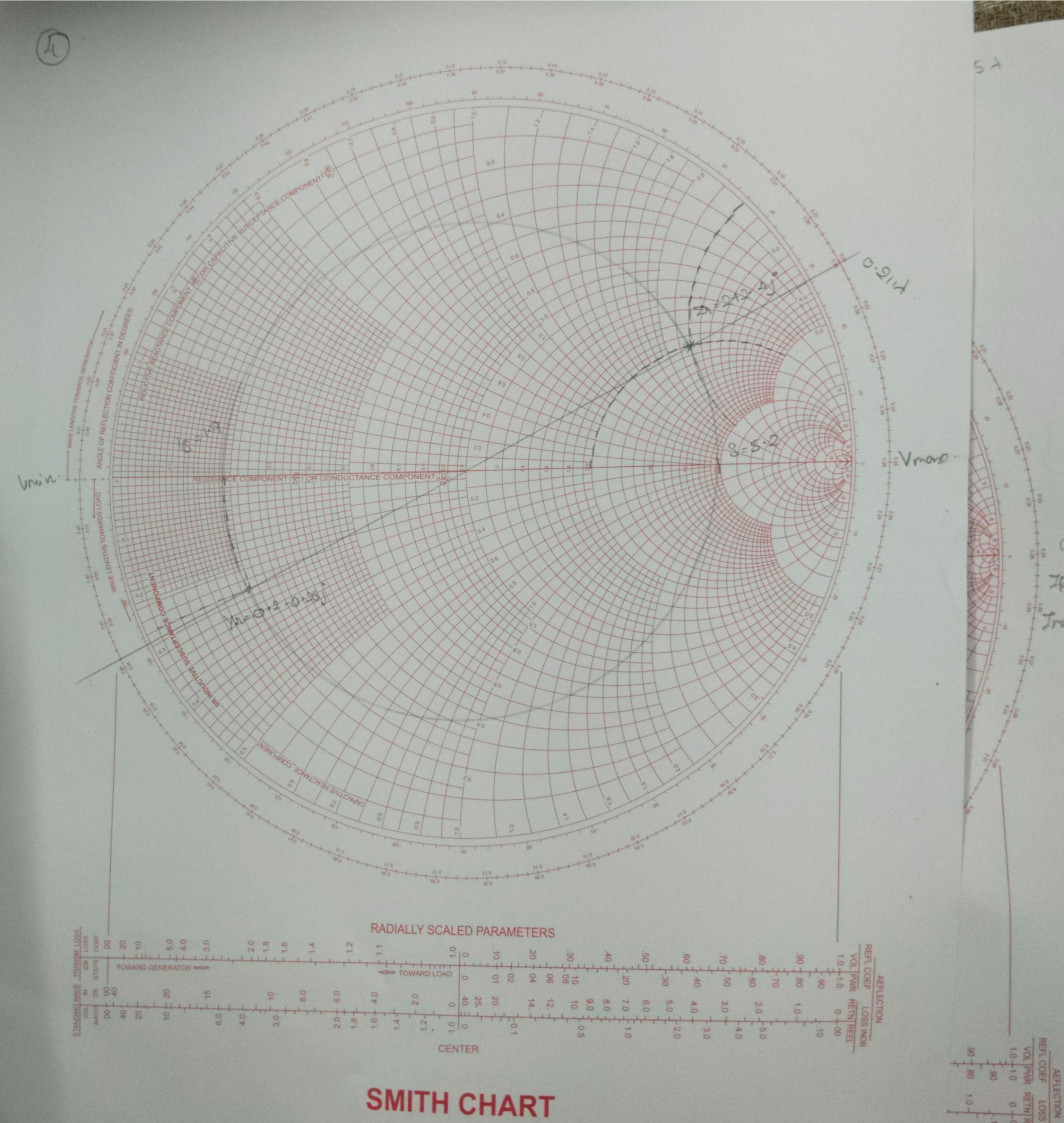
→ half of full circle → here full circle is 0.5. they ask only 1st half max so, $\frac{0.5}{2} = 0.25$

$= 0.25\lambda - 0.211\lambda$

$= 0.039\lambda \quad \therefore \lambda = 2.5m$

$= 0.039\,(2.5)$

$= 0.095$.



RESISTANCE COMPONENT (R/Zo) OR CONDUCTANCE COMPONENT (G/Yo)
INDUCTIVE REACTANCE COMPONENT (+jX/Zo) OR CAPACITIVE SUSCEPTANCE COMPONENT (+jB/Yo)
CAPACITIVE REACTANCE COMPONENT (-jX/Zo) OR INDUCTIVE SUSCEPTANCE COMPONENT (-jB/Yo)
WAVE LENGTHS TOWARDS GENERATOR
WAVE LENGTHS TOWARDS LOAD
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
Vmin
Vmax
0.21λ
S=5.2
RADIALLY SCALED PARAMETERS
TOWARD GENERATOR
TOWARD LOAD
CENTER
AEFLECTION
REFL COEF
LOSS INDB
VOL
PWR
RETN
REEL
STANDING WAVE
TRANSM. LOSS

SMITH CHART

5] Determine the input impedance and standing wave ratio for a distance 1.25 λ long transmission line at the sending end with a characteristic impedance 50 Ω and load impedance $Z_R = 30+40j$ Ω.

Given, $Z_0 = 50\,\Omega$

$Z_R = 30 + 40j$

Soln: Normalized load impedance,

$$Z_n = \frac{Z_R}{Z_0} = \frac{30+40j}{50}$$

$$= \frac{50\angle 53.13}{50}$$

$$= 1\angle 53.13$$

$$Z_n = 0.6 + 0.8j$$

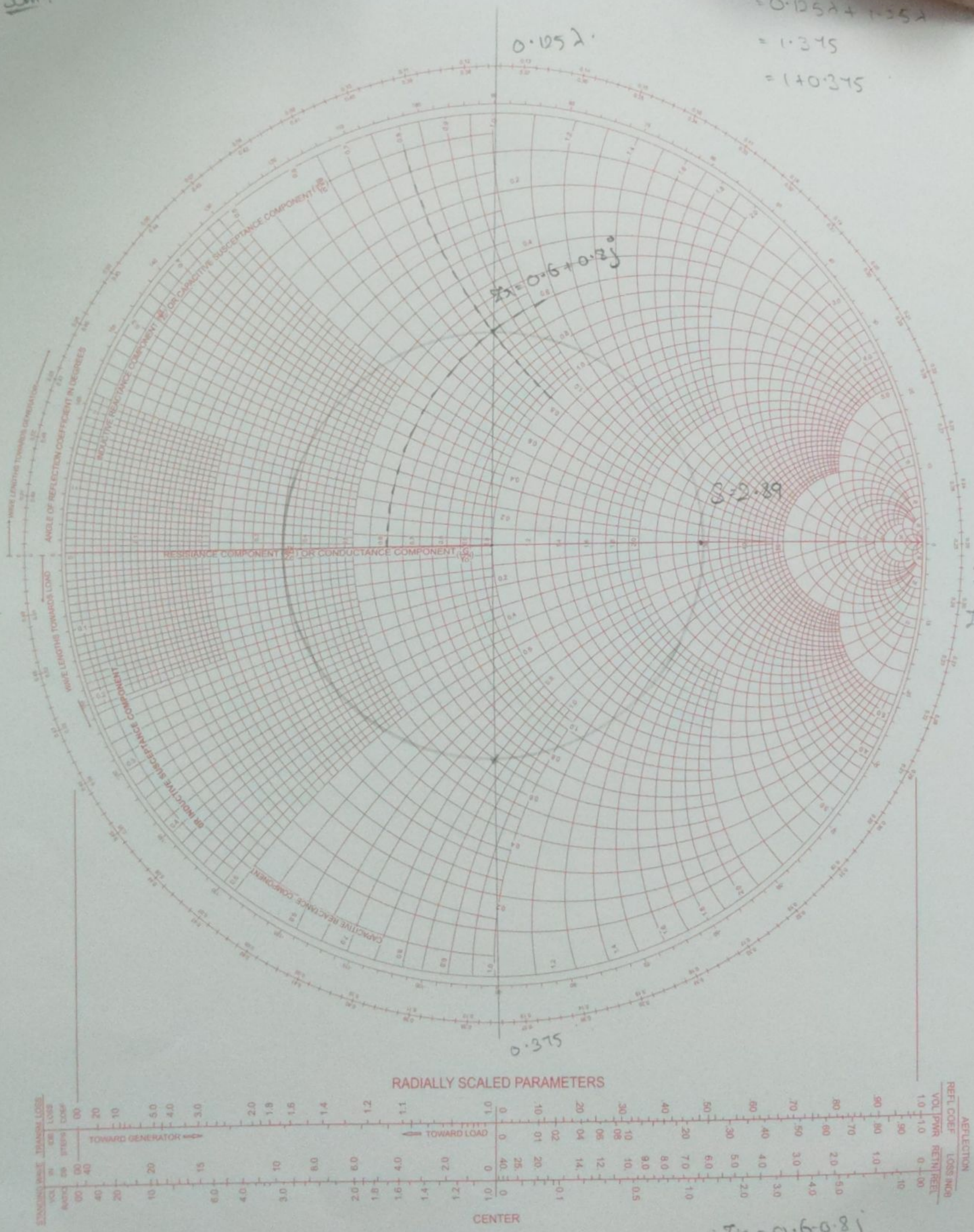
i] SWR = 2.89, ii] $Z_{in} = 0.6 - 0.8j$. $Z_{IN} = 30 - 40j$

(5) Soln..

$= 0.125\lambda + 1.25\lambda$

$= 1.375$

$= 1 + 0.375$



$\therefore Z_{in} = 0.6 - 0.8j$

$Z_{IN} = Z_{in} Z_0 = (0.6 - 0.8j)(50)$

$= 30 - 40j$

SMITH CHART

6 A 75 Ω lossless transmission line is to be matched with a 100-j80 Ω load using single stub. Calculate stub length and its distance from the load correspond to the frequency of 30 MHz using Smith chart.

Soln:

$Z_0 = 75\,\Omega$

$Z_L = 100 - j80\,\Omega$

$f = 30\,MHz = 30 \times 10^6$

$Z_L = Z_0 \, z_l$

$z_l = \frac{Z_L}{Z_0}$

$= \frac{100 - 80j}{75}$

Normalized load impedance,

$$\boxed{z_l = 1.33 - 1.06j}$$

$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6}$

$= 10 \times 10^2 = 10\,m$

$= 1000\,cm$

$Y_\lambda = \frac{1}{1.334 - 1.067j} = 0.457 + 0.365j$

$\downarrow$

$1/z_\lambda$

$$\boxed{l_s = 0.096\lambda}$$

$= 0.096 \times 10$

$l_s = 0.96\,m$

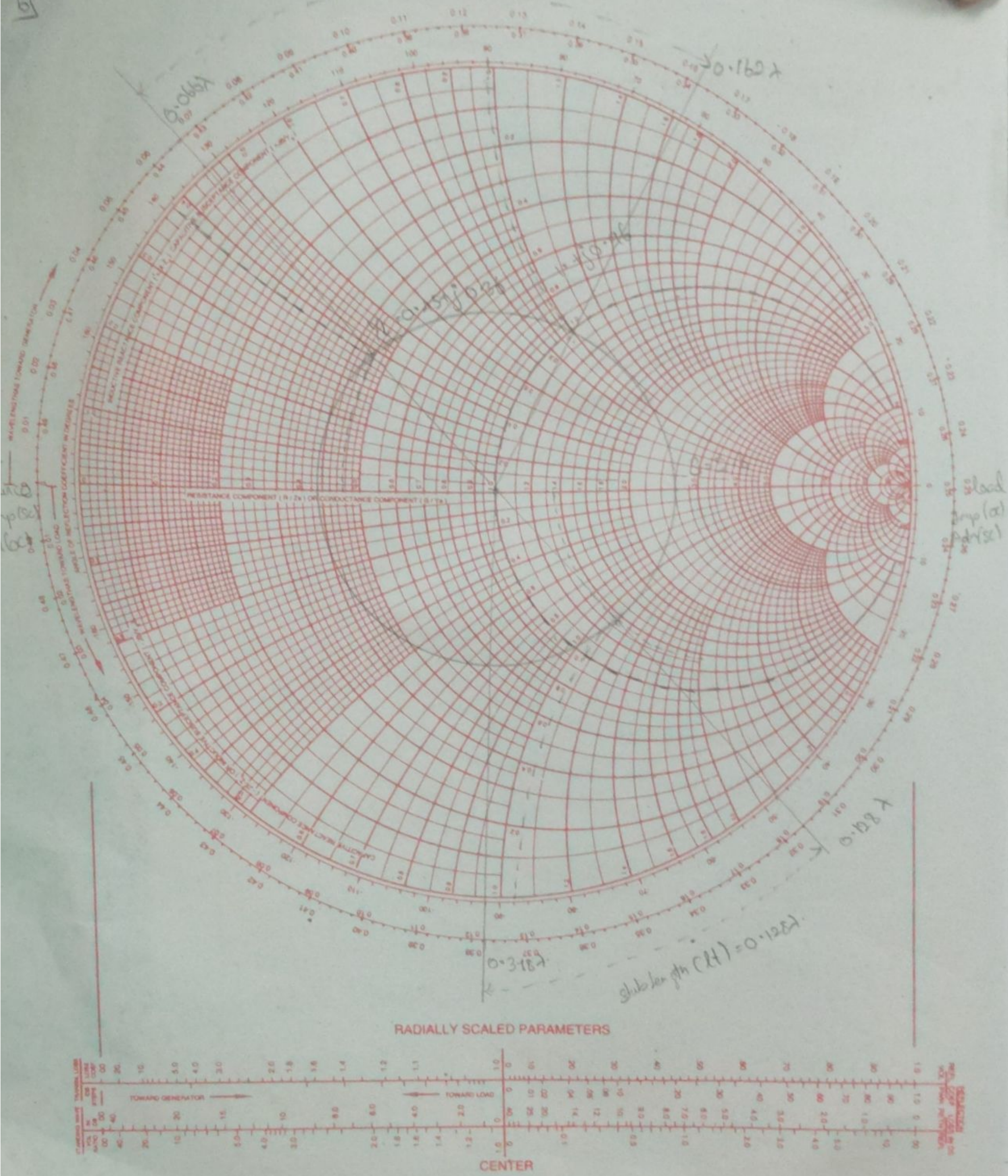
$$l_t = 0.128\lambda$$

$$= 0.128 \times 10$$

$$\boxed{l_t = 1.28\,m}$$

b

distance(lb) = 0.0967λ



7. For a transmission line with $Z_0 = 75\,\Omega$ and a load with the capacitive impedance $Z_L = 45 - j100\,\Omega$. Determine SWR, stub length, stub location by using Smith chart.

Soln:

$$Z_0 = 75\,\Omega$$

$$Z_L = 45 - j100\,\Omega$$

$$Z_L = Z_0 Z_l$$

$$Z_l = \frac{Z_L}{Z_0}$$

$$= \frac{45 - j100}{75}$$

$$\boxed{Z_l = 0.6 - 1.333j}$$

$$y_l = \frac{1}{Z_l} = \frac{1}{0.6 - 1.33j} = 0.281 + 0.624j$$

Stub location:

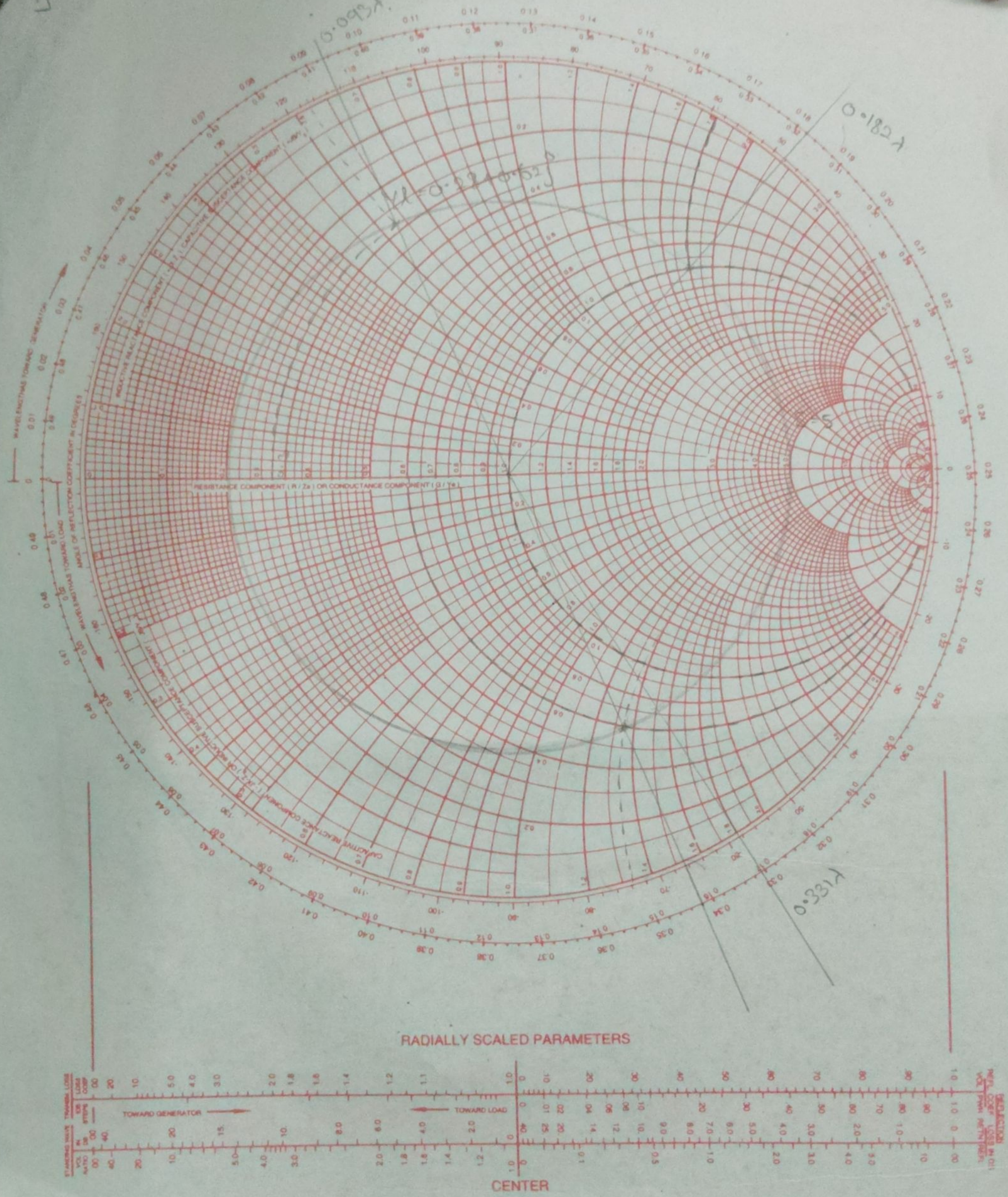
$$l_s = 0.182\lambda - 0.093\lambda$$

$$\boxed{l_s = 0.089\lambda}$$

Stub length:

$$l_t = 0.331\lambda - 0.25\lambda$$

$$\boxed{l_t = 0.081\lambda}$$


0.093λ
0.182λ
0.331λ
RESISTANCE COMPONENT (R/Zo) OR CONDUCTANCE COMPONENT (G/Yo)
WAVELENGTHS TOWARD GENERATOR
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
WAVELENGTHS TOWARD LOAD
RADIALLY SCALED PARAMETERS
TOWARD GENERATOR
TOWARD LOAD
CENTER

SMITH CHART

8. For a transmission line with 'Z_0' = 300 Ω & a load with a complex impedance Z_L = 390 + j600 Ω. Determine SWR, stub location, stub length for a single stub matching using smith chart.

Soln:

$Z_0 = 300\ \Omega$

$Z_L = 390 + j600\ \Omega$

$Z_L = Z_0 z_l$

$z_l = \frac{Z_L}{Z_0} = \frac{390 + j600}{300} = 1.3 + 2j$

$y_l = \frac{1}{z_l} = \frac{1}{1.3 + 2j}$

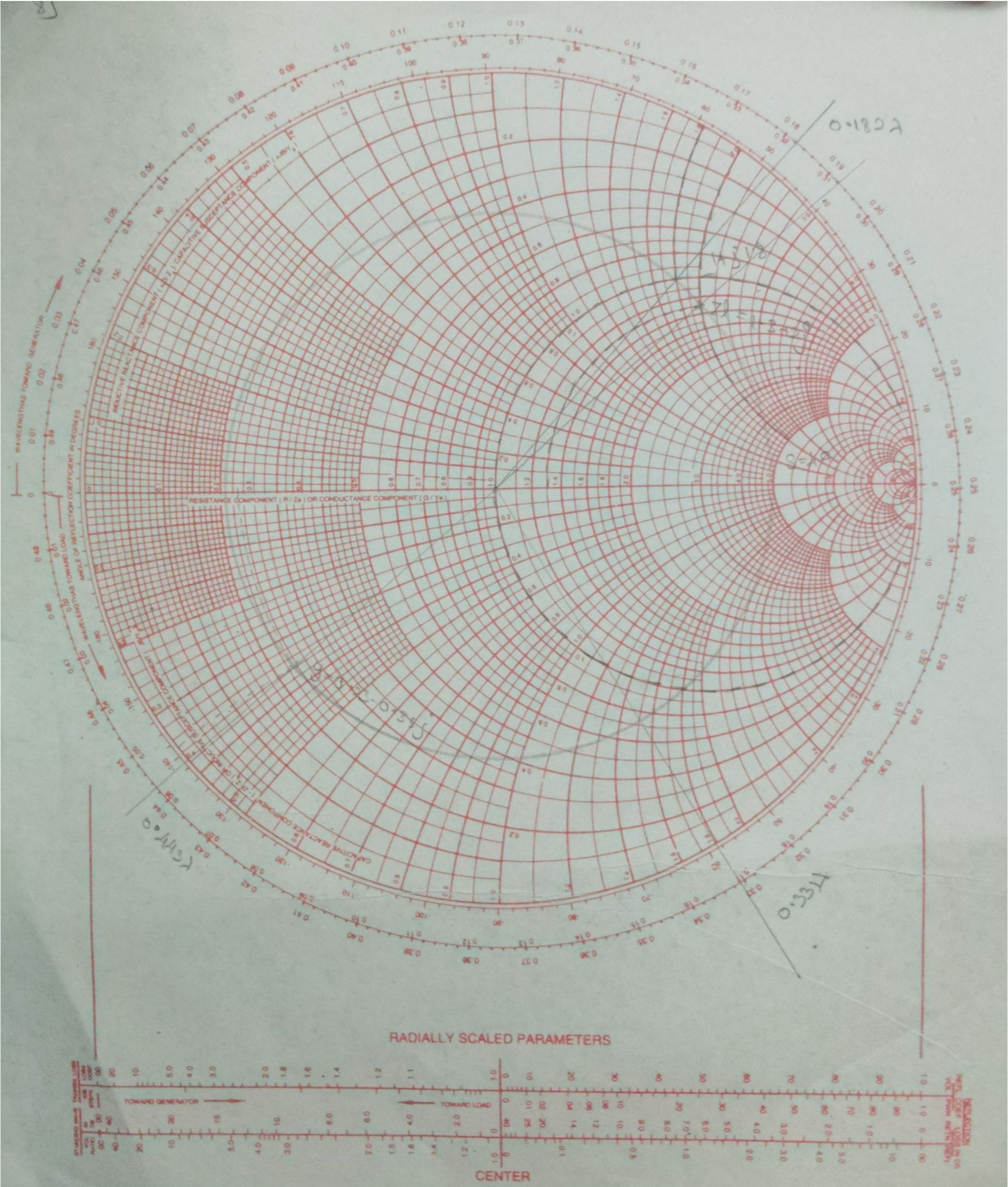
$= 0.228 - 0.351j$

Stub location:

$l_s = 0.261\lambda$

Stub length:

$l_t = 0.081\lambda$.


WAVELENGTHS TOWARD GENERATOR
WAVELENGTHS TOWARD LOAD
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
RESISTANCE COMPONENT (R/Zo), OR CONDUCTANCE COMPONENT (G/Yo)
RADIALLY SCALED PARAMETERS
TOWARD GENERATOR
TOWARD LOAD
CENTER
SMITH CHART